

# Incident Report

|                     |                                                                  |                 |                                                   |
|---------------------|------------------------------------------------------------------|-----------------|---------------------------------------------------|
| Device ID:          | DEV-001                                                          | Sector:         | 1                                                 |
| Device Name:        | AEGIS-PUMP-01                                                    | Timestamp:      | 2026-05-09 09:23:58                               |
| Incident Signature: | c66a4d78f36f7359c7f1929cf8481cc0a89b13b0b203605c8d11048c8cceb5f0 |                 |                                                   |
| Trust Score:        | 14.9%                                                            | Attack Pattern: | Data Exfiltration (large payloads + high entropy) |

## Executive Summary

A critical anomaly was detected on the digital twin for the device. The system automatically generated this report to capture a snapshot of the observed behavior, analysis results, and recommended next steps for investigation and remediation.

## Digital Twin Analysis

The LSTM autoencoder measured the reconstruction error and compared the incoming traffic features to the expected baseline. These discrepancies were used to compute a Trust Score and to highlight the most anomalous data attributes.

| Metric                     | Value   |
|----------------------------|---------|
| Trust Score                | 14.9%   |
| Reconstruction Error (MSE) | 0.47143 |
| Jensen-Shannon Divergence  | 0.40801 |

## Top Anomaly Contributing Features

| Feature     | Delta |
|-------------|-------|
| Entropy     | 0.700 |
| Packet Size | 0.600 |
| Symmetry    | 0.600 |

## Behavioral Timeline

- 14:53:57: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 14:53:52: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 14:53:47: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 14:53:42: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 14:53:37: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 14:53:32: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 14:53:26: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 14:53:23: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471

14:53:19: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471

14:53:15: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471

Possible Attack Pattern

Data Exfiltration (large payloads + high entropy)

Risk Assessment

Based on the observed anomaly magnitude and the estimated severity, this incident is considered **CRITICAL**. The device should be treated as compromised until a full investigation confirms otherwise.

Recommended Remediation Actions

- Isolate the device from the network (quarantine).
- Collect full packet captures and system logs for post-incident analysis.
- Perform a firmware integrity check and verify configuration drift.
- Rotate credentials and revoke all active sessions for the device.
- Update IDS/IPS rules to detect similar behavior in the future.

Digital Evidence Snapshot

| Baseline Features          | Current Features           |
|----------------------------|----------------------------|
| 0.400, 0.500, 0.300, 0.600 | 1.000, 0.000, 1.000, 0.000 |

Incident Signature

c66a4d78f36f7359c7f1929cf8481ccde899613b0b203605cada04e8ccebc5f0